OAS-007 — Memory Architecture

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000
- OAS-001
- OAS-002
- OAS-003
- OAS-004
- OAS-005
- OAS-006
- OCS-000 through OCS-032

Purpose

This specification defines the architectural organization of Memory.

The Constitution defines what Memory is.

This specification defines how Memory is organized so that constitutional continuity survives indefinitely.

Architecture SHALL preserve Memory.

Storage SHALL implement Memory.

The two SHALL remain distinct.

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1. Architectural Objective

The objective of Memory Architecture is to ensure that no constitutional understanding becomes permanently unrecoverable.

Architecture SHALL preserve:

Identity.

Meaning.

Relationships.

History.

Context.

Reasoning.

Memory SHALL remain reconstructable.

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2. Architectural Layers of Memory

Memory SHALL be organized into complementary architectural layers.

At minimum:

Working Memory

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Place Memory

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Library Memory

↓

Federated Memory

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Constitutional Memory

Each layer SHALL preserve the layers beneath it.

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3. Working Memory

Working Memory represents active constitutional interaction.

Working Memory MAY evolve rapidly.

Working Memory SHALL remain connected to permanent constitutional history.

Working Memory SHALL never become isolated.

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4. Place Memory

Every Place SHALL preserve its own architectural memory.

Place Memory SHALL include:

Participants.

Voices.

Questions.

Evidence.

Experiments.

Understanding.

Decisions.

Historical evolution.

Place Memory SHALL remain sovereign.

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5. Library Memory

Library Memory SHALL organize reusable constitutional understanding.

Organization SHALL occur through:

Relationships.

Meaning.

History.

Context.

Library Memory SHALL maximize future discoverability.

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6. Federated Memory

Independent Places MAY contribute to Federated Memory.

Federation SHALL preserve:

Local ownership.

Local history.

Shared discoverability.

Independent reconstruction.

Memory federation SHALL avoid centralized dependency.

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7. Constitutional Memory

Constitutional Memory represents the architectural preservation of the discipline itself.

Constitutional Memory SHALL preserve:

Foundations.

Architecture.

Implementations.

Historical evolution.

Canonical definitions.

Constitutional continuity.

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8. Memory Relationships

Every architectural memory layer SHALL preserve explicit relationships.

Relationships SHALL remain:

Traceable.

Explainable.

Historically attributable.

Relationship preservation SHALL take precedence over storage optimization.

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9. Memory Migration

Architectural Memory SHALL survive:

Technology replacement.

Repository migration.

Implementation redesign.

Organizational change.

Artificial intelligence replacement.

Migration SHALL preserve constitutional continuity.

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10. Memory Compression

Implementations MAY compress representation.

Architecture SHALL preserve reconstructability.

Compression SHALL NEVER eliminate constitutional meaning.

Optimization SHALL remain subordinate to constitutional continuity.

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11. Memory Recovery

Architectural Memory SHALL support recovery using only:

Constitutional identity.

Architectural relationships.

Historical lineage.

Applicable constitutional specifications.

Recovery SHALL remain implementation-independent.

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12. Memory Growth

Memory SHALL grow through constitutional evolution.

Growth SHALL improve:

Coverage.

Connectivity.

Historical richness.

Explainability.

Memory SHALL remain coherent despite indefinite expansion.

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13. Architectural Pattern

Memory Architecture SHALL organize systems around preserved relationships rather than accumulated records.

Records preserve events.

Relationships preserve understanding.

Architecture SHALL preserve both.

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14. Architectural Verification

Verification SHALL demonstrate that constitutional memory remains reconstructable after arbitrary implementation replacement.

Equivalent storage SHALL NOT be required.

Equivalent Memory SHALL.

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15. Architectural Memory Test

Memory Architecture satisfies this specification when an independent architect can reconstruct:

What was remembered.

Why it was remembered.

How it relates to everything else.

How understanding evolved.

How memory survives implementation loss.

How future generations continue extending it.

If any answer cannot be reconstructed,

Memory Architecture is incomplete.

Conformance

An implementation conforms to this specification when Memory remains constitutionally continuous, architecturally reconstructable, and semantically preserved independently of storage technology.

Architectural Principle

Storage remembers bytes.

Databases remember records.

Backups remember files.

Memory remembers meaning.

Architecture therefore exists not to preserve data,

but to preserve the living continuity through which every future generation can inherit, understand, question, and extend the constitutional memory of Organodynamics.

End of Specification OAS-007